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Indian Institute of Technology Hyderabad

DSP Lab - EE2801

Experiment-1

BY

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Contents

1	Objective	1
2	Theory	2
3	Procedure	3
4	Code	4
5	Results and Observations	5
6	Conclusion	10

1 Objective

Part A: Perform the following using MATLAB

1. Generate and plot samples of $x(t) = 2 \sin(2\pi ft)$, where $f = 1$ kHz and $F_s = 48$ kHz, for a 1-second duration. (Plot only 5 cycles).
2. Convert the samples to fixed-point formats: $Q(2, 14)$, $Q(4, 12)$, and $Q(8, 4)$.
3. Plot the quantized signals versus the original signal.
4. Plot the errors for each case.
5. Calculate the Signal-to-Quantization-Noise Ratio (SQNR) for each case using:

$$SQNR = \frac{\text{mean}(|x[n]|^2)}{\text{mean}(|e[n]|^2)}$$

Part B: Using Verilog

1. Generate samples of $x(t) = 2 \sin(2\pi ft)$, where $f = 1$ kHz and $F_s = 48$ kHz, for a 1-second duration.
2. Convert the samples to fixed-point formats: $Q(2, 14)$, $Q(4, 12)$, and $Q(8, 4)$.
3. Write the quantized signals, original signal, and errors to a file (or files).
4. Calculate the SQNR for each case using:

$$SQNR = \frac{\text{mean}(|x[n]|^2)}{\text{mean}(|e[n]|^2)}$$

5. Plot the quantized signals versus the original signal in MATLAB using the exported file data. Verify that the plots from the Verilog data match the MATLAB-generated plots.
6. Repeat step 5 for the error signals.

2 Theory

A continuous-time sinusoidal signal is given by

$$x(t) = 2 \sin(2\pi ft)$$

Sampling at a frequency F_s converts it into a discrete-time signal

$$x[n] = 2 \sin\left(2\pi \frac{f}{F_s} n\right)$$

where $T_s = \frac{1}{F_s}$. For $f = 1$ kHz and $F_s = 48$ kHz, one cycle contains 48 samples.

Fixed-Point Representation

In digital systems, fixed-point arithmetic is commonly used. A number in $Q(m, n)$ format consists of m integer bits, including the sign bit and n fractional bits. Quantization is performed as

$$x_q[n] = \frac{\text{fix}(x[n] \cdot 2^n)}{2^n}$$

The formats $Q(2, 14)$, $Q(4, 12)$, and $Q(8, 4)$ are used.

Quantization Error

Quantization introduces an error given by

$$e[n] = x[n] - x_q[n]$$

Signal-to-Quantization Noise Ratio

The Signal-to-Quantization Noise Ratio (SQNR) is defined as

$$\text{SQNR} = \frac{\text{mean}(|x[n]|^2)}{\text{mean}(|e[n]|^2)}$$

and in decibels as

$$\text{SQNR}_{\text{dB}} = 10 \log_{10}(\text{SQNR})$$

Verilog Implementation

A lookup table (LUT) is used in Verilog to generate one cycle of the sine wave, which is repeated to obtain the required number of samples. MATLAB is used to verify the Verilog-generated results.

3 Procedure

Part A: MATLAB

- Generate a sinusoidal signal $x(t) = 2 \sin(2\pi ft)$ with $f = 1$ kHz.
- Sample the signal at $F_s = 48$ kHz for a duration of one second.
- Plot only five cycles of the sampled signal.
- Convert the samples into fixed-point formats $Q(2, 14)$, $Q(4, 12)$, and $Q(8, 4)$.
- Plot the quantized signals along with the original signal.
- Compute and plot the quantization error for each format.
- Calculate the SQNR for each fixed-point representation.

Part B: Verilog

- Create a lookup table (LUT) to store one cycle of the sine wave.
- Generate 48,000 samples corresponding to one second of the signal.
- Quantize the samples into $Q(2, 14)$, $Q(4, 12)$, and $Q(8, 4)$ formats.
- Compute the quantization error for each format.
- Write the original signal, quantized signals, and error values to text files.

MATLAB Verification

- Import the Verilog-generated data files into MATLAB.
- Plot the original and quantized signals using the file data.
- Plot the quantization errors for each fixed-point format.
- Compute the SQNR and compare the results with MATLAB-only implementation.

4 Code

MATLAB Code – Part A (Core Logic)

The following MATLAB code shows the core logic for signal generation, fixed-point quantization, and SQNR computation.

```
x = 2*sin(2*pi*f*t);

x_2_14 = fix(x*2^14)/2^14;
x_4_12 = fix(x*2^12)/2^12;
x_8_4  = fix(x*2^4)/2^4;

error_2_14 = x - x_2_14;
error_4_12 = x - x_4_12;
error_8_4  = x - x_8_4;

SQNR_2_14 = mean(abs(x).^2)/mean(abs(error_2_14).^2);
SQNR_4_12 = mean(abs(x).^2)/mean(abs(error_4_12).^2);
SQNR_8_4  = mean(abs(x).^2)/mean(abs(error_8_4).^2);
```

Verilog Code – Part B (Core Logic)

The following Verilog snippet illustrates sine generation using a lookup table, fixed-point quantization, and error computation.

```
x = sine_lut[n % 48];

q214 = $rtoi(x * (1 << 14));
x_2_14 = q214 / (1.0 * (1 << 14));
e_2_14 = x - x_2_14;

$fwrite(f_x, "%f\n", x);
$fwrite(f_q214, "%f\n", x_2_14);
$fwrite(f_e214, "%f\n", e_2_14);
```

The complete MATLAB and Verilog source codes are provided in the Appendix.

5 Results and Observations

The sinusoidal signal $x(t) = 2 \sin(2\pi ft)$ with $f = 1$ kHz was sampled at $F_s = 48$ kHz for one second. Only five cycles are shown in all plots for clarity.

MATLAB Results

The signal was generated and quantized in MATLAB using fixed-point formats $Q(2, 14)$, $Q(4, 12)$, and $Q(8, 4)$.

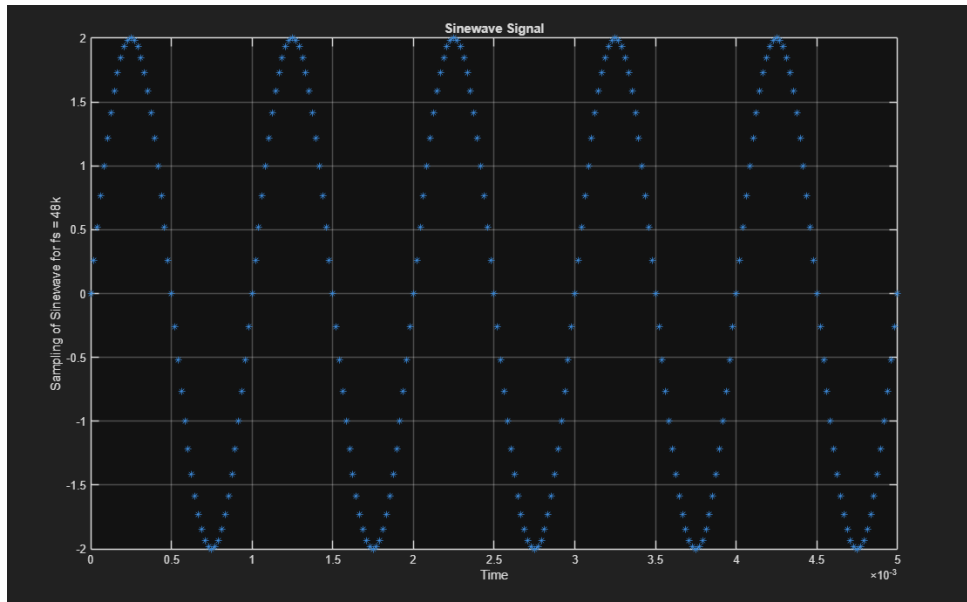


Figure 1: MATLAB: Sampled sine wave (5 cycles)

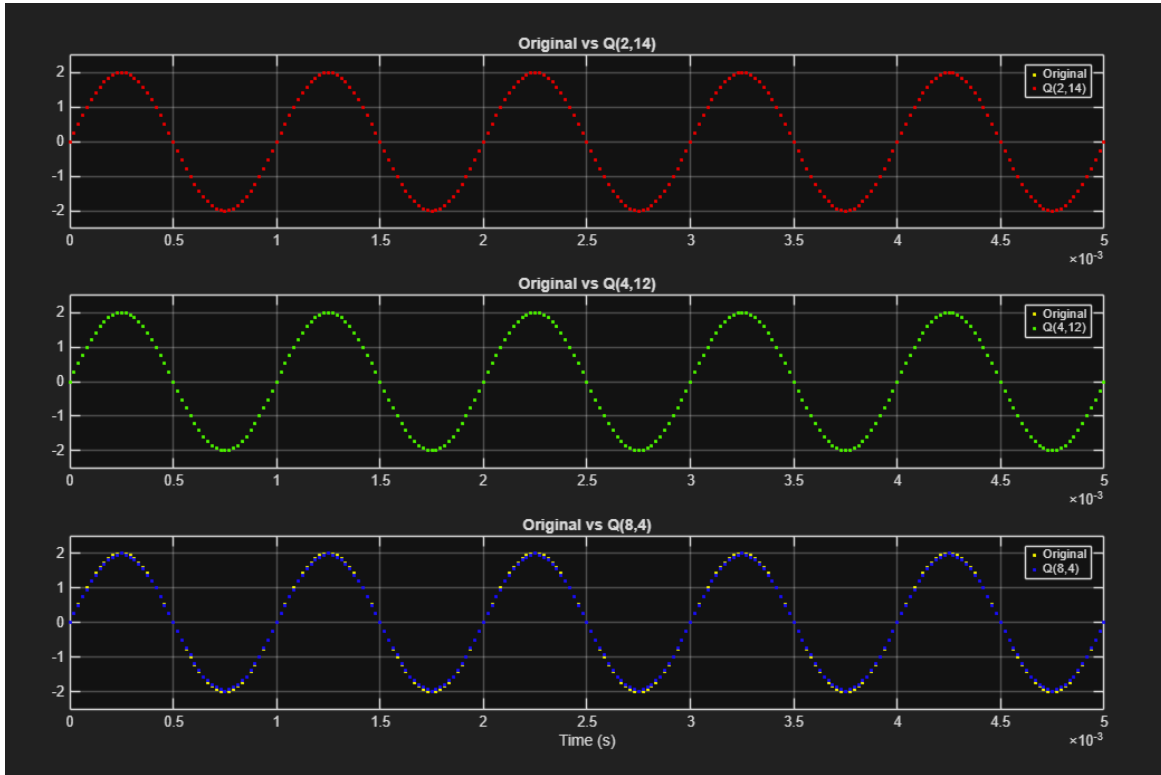


Figure 2: MATLAB: Original vs quantized signals

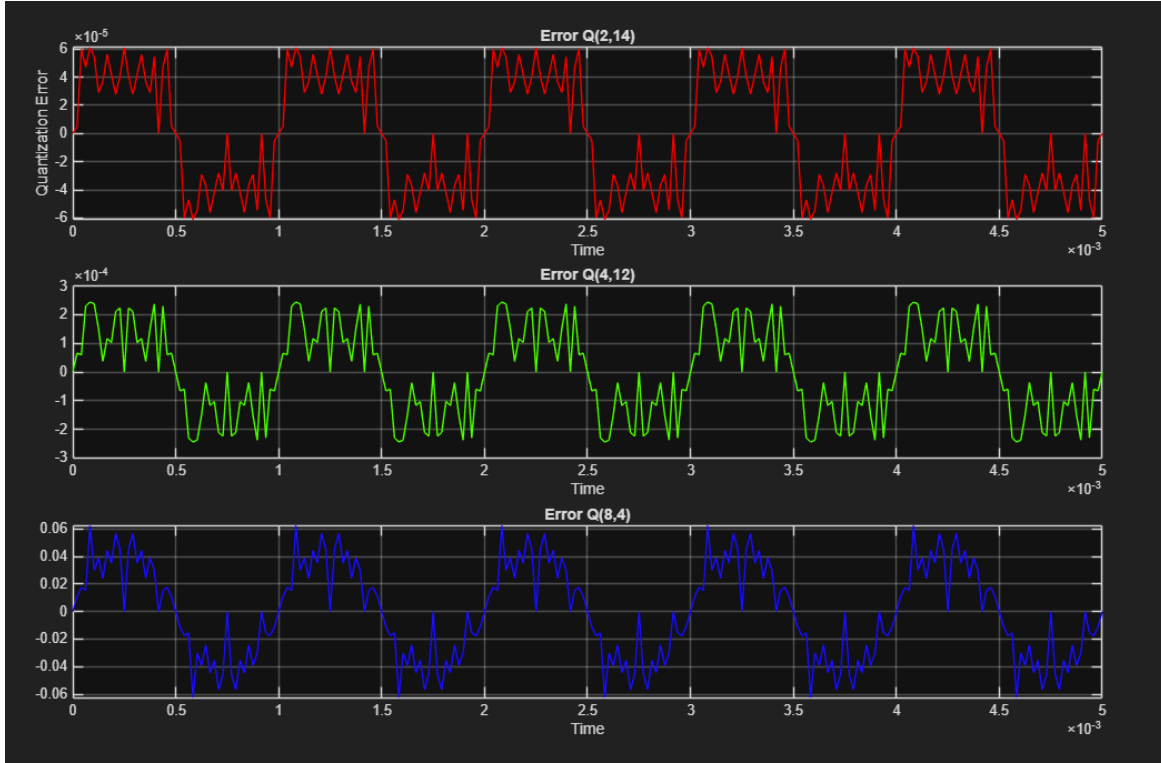


Figure 3: MATLAB: Quantization errors

It is observed that $Q(2, 14)$ closely matches the original signal, while $Q(8, 4)$ shows significant quantization distortion.

Verilog Results

The same signal was generated in Verilog using a lookup table and quantized into the same fixed-point formats. The Verilog-generated data was plotted using MATLAB.

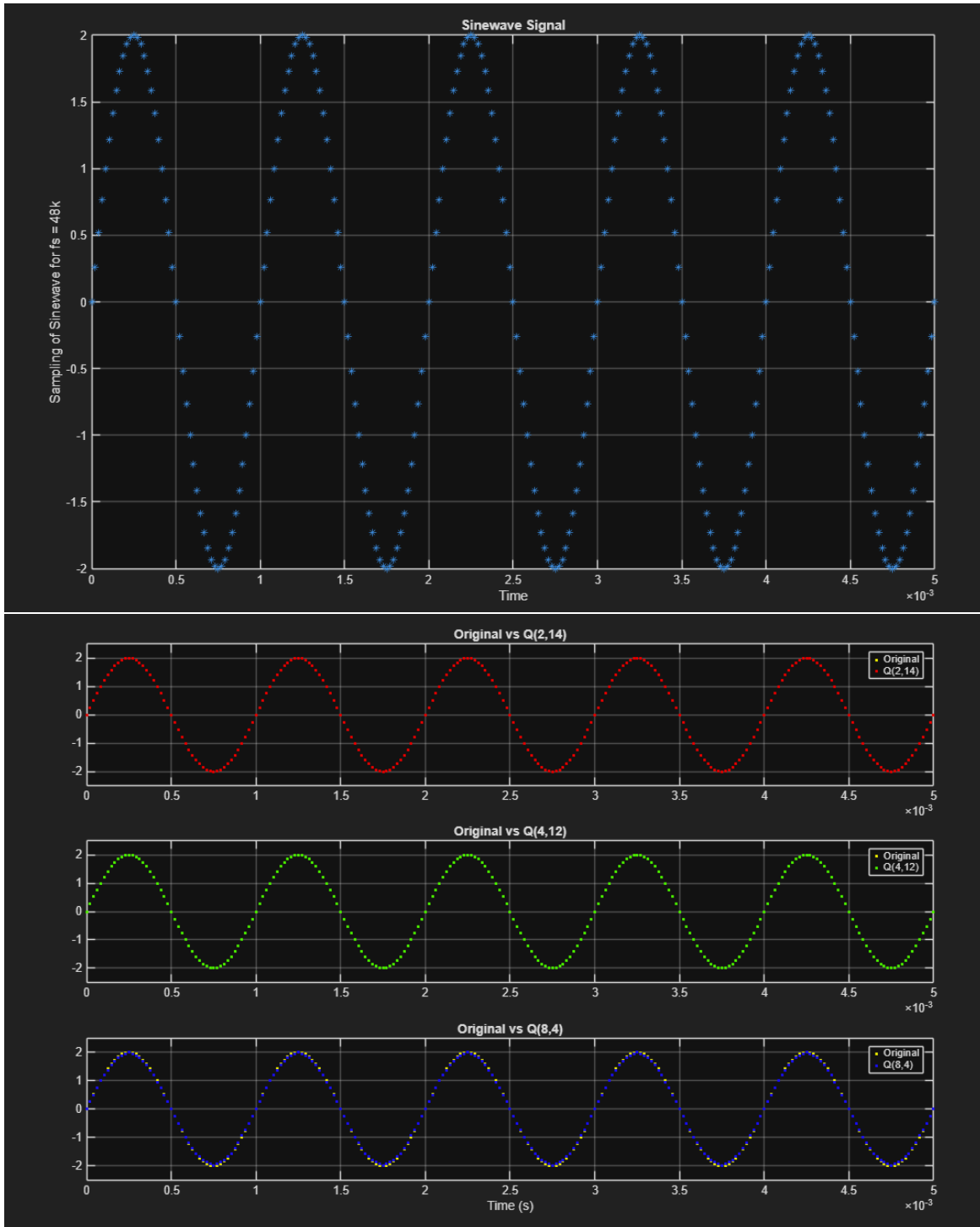


Figure 4: Verilog: Original vs quantized signals

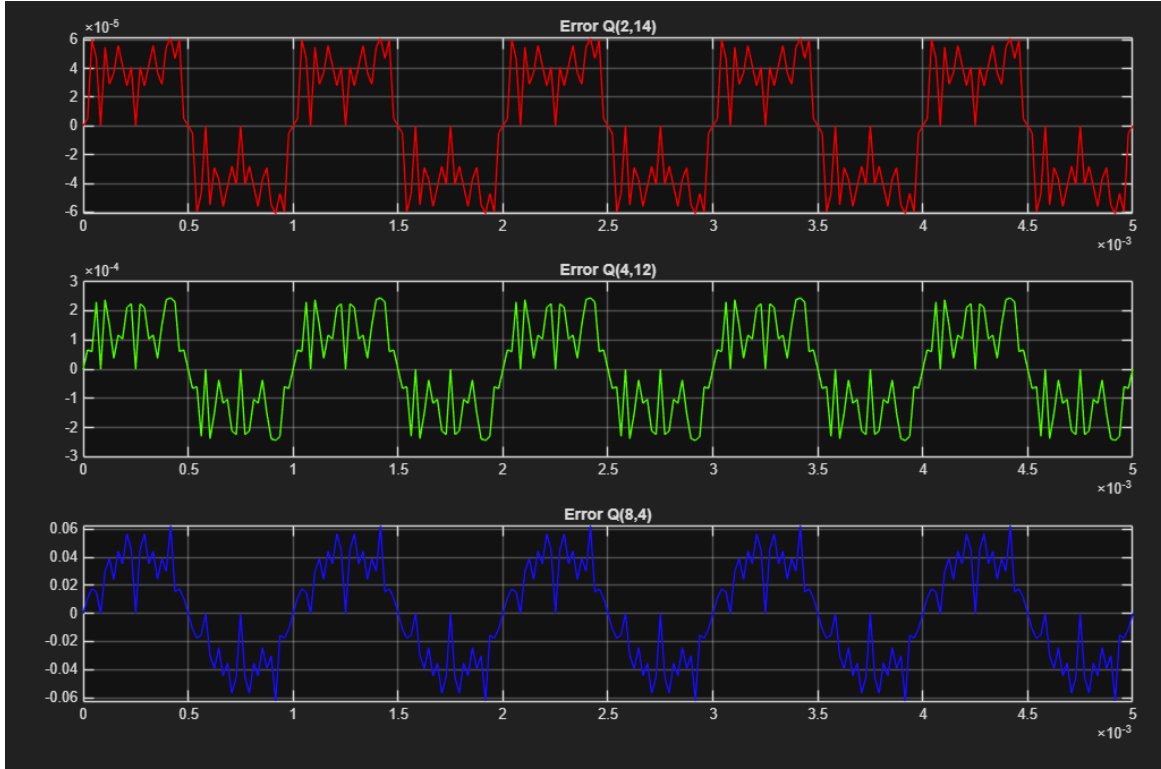


Figure 5: Verilog: Quantization errors

The Verilog plots match the MATLAB plots, confirming correct fixed-point implementation. The SQNR was highest for $Q(2, 14)$ and lowest for $Q(8, 4)$, indicating that quantization noise increases as the number of fractional bits decreases.

Q-Format	SQNR (MATLAB) [dB]	SQNR (Verilog) [dB]
Q(2,14)	90.55	90.73
Q(4,12)	79.17	79.17
Q(8,4)	32.29	32.29

Table 1: Comparison of SQNR values obtained from MATLAB and Verilog implementations

6 Conclusion

In this experiment, a sinusoidal signal was generated and quantized using different fixed-point formats in MATLAB and Verilog. The effect of fractional bit width on quantization error and SQNR was analyzed. It was observed that higher fractional precision results in lower quantization error and higher SQNR. The SQNR values obtained from Verilog closely match those from MATLAB, confirming the correctness of the Verilog implementation and fixed-point modeling.

Source Code Repository:

The complete MATLAB and Verilog source codes used in this experiment are available at the following GitHub repository:

<https://github.com/Koushik-11044/DSP>